

Chih-Yuan Chiu

ACADEMIC EMPLOYMENT	Georgia Institute of Technology <i>Research Engineer II,</i> <i>School of Electrical and Computer Engineering.</i> <i>Supervisor: Professor Justin Romberg</i>	Aug 2024 - now
EDUCATION	University of California, Berkeley <i>Ph.D., Department of Electrical Engineering and Computer Sciences</i> <i>Advisor: Professor Shankar Sastry</i> National Taiwan University <i>B.S., Department of Electrical Engineering</i>	Sept 2018 - Dec 2023 Sept 2014 - June 2018
CONTACT INFORMATION	<i>Emails:</i> cyc@gatech.edu, chihyuanfrankchiu@gmail.com. <i>Website:</i> https://chihyuanchiu.github.io/ <i>Google Scholar:</i> https://scholar.google.com/citations?hl=en&user=cl9ModoAAAAJ	
PREPRINTS	Jihoon Hong, Chih-Yuan Chiu , Sara Fridovich-Keil, Glen Chou. “PolyMerge: Compressing 3D Gaussian Splats with Polytope Coverings for Provably Safe Resource-Constrained Navigation.” (Submitted to: <i>IEEE Robotics and Automation Letters (RA-L)</i> .) Dan BW Choe, Sundhar Vinodh Sangeetha, Steven Emanuel, Chih-Yuan Chiu , Samuel Coogan, Shreyas Kousik. “Ask, Reason, Assist: Decentralized Robot Collaboration via Language and Logic,” ArXiv preprint 2509.23506, September 2025. URL: https://arxiv.org/pdf/2509.23506 . Zhouyu Zhang ¹ , Chih-Yuan Chiu *, Glen Chou. “Constraint Learning in Multi-Agent Dynamic Games from Demonstrations of Local Nash Interactions,” ArXiv preprint 2508.19945, August 2025. URL: https://arxiv.org/abs/2508.19945 . (Revise and Resubmit at: <i>IEEE Robotics and Automation Letters (RA-L)</i> .)	
JOURNAL PUBLICATIONS	Chih-Yuan Chiu *, Zhouyu Zhang*, Glen Chou. “Learning Constraints from Stochastic Partially-Observed Closed-Loop Demonstrations,” ArXiv preprint 2509.15109, September 2025. URL: https://arxiv.org/abs/2509.15109 . (Accepted at: <i>IEEE Control Systems Letters (L-CSS)</i> .) Chih-Yuan Chiu , Bryce Ferguson. “Robustness of Incentive Mechanisms Against System Misspecification in Congestion Games,” in <i>IEEE Control Systems Letters</i> , vol. 9, pp. 276-281, May 2025. Chih-Yuan Chiu *, Jingqi Li*, Maulik Bhatt, Negar Mehr. “To What Extent Do Open-Loop and Feedback Nash Equilibria Diverge in General-Sum Linear Quadratic Dynamic Games?” <i>IEEE Control Systems Letters (L-CSS)</i> , vol. 8, pp. 2583-2588, November 2024. Lasse Peters, Andrea Bajcsy, Chih-Yuan Chiu , David Fridovich-Keil, Forrest Laine, Laura Fer-ranti, Javier Alonso-Mora. “Contingency Games for Multi-Agent Interaction,” <i>IEEE Robotics and Automation Letters (RA-L)</i> , vol. 9, pp. 2208-2215, January 2024. Forrest Laine, David Fridovich-Keil, Chih-Yuan Chiu , and Claire Tomlin. “The Computation of Approximate Generalized Feedback Nash Equilibria”, <i>SIAM Journal on Optimization</i> , vol. 33, no. 1, pp. 294-318, 2023. Druv Pai, Michael Psenka, Chih-Yuan Chiu , Manxi Wu, Edgar Dobriban, Yi Ma. “Pursuit of a Discriminative Representation for Multiple Subspaces via Sequential Games”, vol. 360, no. 6, <i>Journal of the Franklin Institute</i> , pp. 4135-4171, April 2023.	

¹*Equal contribution.

Amay Saxena*, **Chih-Yuan Chiu***, Ritika Shrivastava, Joseph Menke, Shankar Sastry. “Simultaneous Localization and Mapping: Through the Lens of Nonlinear Optimization,” vol. 7, no. 3, *IEEE Robotics and Automation Letters (RA-L)*, July 2022.

Shuyu Zhan*, **Chih-Yuan Chiu***, Antoine Leeman, Glen Chou. “Robustly Constrained Dynamic Games for Uncertain Nonlinear Dynamics,” ArXiv preprint 2509.16826, September 2025. URL: <https://arxiv.org/abs/2509.16826>. Accepted at *2026 IEEE International Conference on Robotics and Automation (ICRA)*.

Sundhar Vinodh Sangeetha, **Chih-Yuan Chiu**, Sarah H.Q. Li, Shreyas Kousik “Language Conditioning Improves Accuracy of Aircraft Goal Prediction in Untowered Airspace,” ArXiv preprint 2509.14063, September 2025. URL: <https://arxiv.org/abs/2509.14063>. Accepted at *2026 IEEE International Conference on Robotics and Automation (ICRA)*.

Zheng Qiu, **Chih-Yuan Chiu**, Glen Chou. “Active Constraint Learning in High Dimensions from Demonstrations,” ArXiv preprint 2512.22757, December 2025. Accepted at: *8th Annual Learning for Dynamics & Control Conference (L4DC)*.

Chih-Yuan Chiu. “Approximately Optimal Toll Design for Efficiency and Equity in Arc-Based Traffic Assignment Models,” *60th Annual Allerton Conference on Communication, Control, and Computing*, September 2025.

Chih-Yuan Chiu, Devansh Jalota, Marco Pavone. “Credit vs. Discount-Based Congestion Pricing: A Comparison Study,” in *IEEE Conference on Decision and Control (CDC)*, pp. 2331-2336, December 2024.

Chih-Yuan Chiu, Shankar Sastry. “Parameter Estimation in Optimal Tolling for Traffic Networks Under the Markovian Traffic Equilibrium,” in *American Control Conference (ACC)*, pp. 1461-1467, July 2024.

Chih-Yuan Chiu, Chinmay Maheshwari, Pan-Yang Su, Shankar Sastry. “Dynamic Tolling in Arc-based Traffic Assignment Models,” in *59th Annual Allerton Conference on Communication, Control, and Computing*, pp. 1-8, September 2023.

Chih-Yuan Chiu*, Chinmay Maheshwari*, Pan-Yang Su, Shankar Sastry. “Arc-based Traffic Assignment: Equilibrium Characterization and Learning,” in *IEEE Conference on Decision and Control (CDC)*, pp. 7751-7758, December 2023.

Jingqi Li, **Chih-Yuan Chiu**, Lasse Peters, Fernando Palafox, Mustafa Karabag, Javier Alonso-Mora, Somayeh Sojoudi, Claire Tomlin, David Fridovich-Keil. “Scenario-Game ADMM: A Parallelized Scenario-Based Solver for Stochastic Noncooperative Games,” in *IEEE Conference on Decision and Control (CDC)*, pp. 8093-8099, December 2023.

Chih-Yuan Chiu. “SLAM Backends with Objects in Motion: A Unifying Framework and Tutorial,” in *American Control Conference (ACC)*, pp. 1635-1642, May 2023.

Jingqi Li, **Chih-Yuan Chiu**, Lasse Peters, Somayeh Sojoudi, Claire Tomlin, David Fridovich-Keil. “Cost Inference for Feedback Dynamic Games from Noisy Partial State Observations and Incomplete Trajectories,” in *International Conference on Autonomous Agents and Multiagent Systems*, pp. 1062-1070, June 2023.

Chih-Yuan Chiu, David Fridovich-Keil. “GTP-SLAM: Game-Theoretic Priors for Simultaneous Localization and Mapping in Multi-Agent Scenarios,” in *IEEE Conference on Decision and Control (CDC)*, pp. 247-252, December 2022.

Chinmay Maheshwari*, **Chih-Yuan Chiu***, Eric Mazumdar, Shankar Sastry and Lillian J. Ratliff. “Zeroth-Order Methods for Convex-Concave Minmax Problems: Applications to Decision-Dependent Risk Minimization”, in *25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 51, pp. 6702-6734, 2022.

Chih-Yuan Chiu*, David Fridovich-Keil*, and Claire Tomlin. “Encoding Defensive Driving as a Dynamic Nash Game,” in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10749-10756, 2021.

Forrest Laine, David Fridovich-Keil, **Chih-Yuan Chiu**, and Claire Tomlin. “Multi-Hypothesis Interactions in Game-Theoretic Motion Planning”, in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 8016-8023, June 2021.

Forrest Laine, **Chih-Yuan Chiu**, Claire Tomlin, “Eyes-Closed Safety Kernels: Safety for Autonomous Systems Under Loss of Observability”, in *Robotics: Science and Systems (RSS)*, July 2020.

THESES

(*Ph.D. Thesis*) **Chih-Yuan Chiu**. “Algorithm Design for Safe and Efficient Societal-Scale Navigation,” 2023. (<https://www2.eecs.berkeley.edu/Pubs/TechRpts/2023/EECS-2023-267.pdf>).

(*Master’s Thesis*) **Chih-Yuan Chiu**. “Simultaneous Localization and Mapping: A Rapprochement of Filtering and Optimization-Based Approaches,” 2021. (<https://www2.eecs.berkeley.edu/Pubs/TechRpts/2021/EECS-2021-76.pdf>).

INVITED ORAL PRESENTATIONS

11th Georgia Tech Postdoctoral Research Symposium Nov 2025
“Credit-based vs. Discount-based Congestion Pricing: A Comparison Study.” (Top 10 Submissions)

Georgia Tech DCL Symposium Spotlight Talk Apr 2025
“Credit-based vs. Discount-based Congestion Pricing: A Study of Tolling and Toll Relief Mechanisms for Traffic Management.”

Georgia Tech ECE Decision and Control Laboratory Student Seminar Oct 2024
“Interaction-aware Multi-Agent Control: The Role of Information.”

UC Berkeley EECS C106B/206B Guest Lecture Apr 2022
“Simultaneous Localization and Mapping: A Unifying Optimization-Based Framework.” (with Ritika Srivastava)

UC Berkeley Semiautonomous Seminar Feb 2022
“Towards a Rapprochement of Estimation, Prediction, and Planning for Autonomous Navigation.”

UC Berkeley EE 221A Guest Lecture Dec 2021
“Simultaneous Localization and Mapping: Filtering and Optimization Approaches.” (with Amay Saxena)

UC Berkeley Semiautonomous Seminar Sep 2021
“Gradient Free Optimistic Gradient Descent Ascent: Applications to Decision Dependent Risk Minimization.” (with Chinmay Maheshwari)

UC Berkeley Semiautonomous Seminar Jun 2021
“Factor Graphs: A Tool for Optimization-Based Inference in Robotics.”

UC Berkeley EECS C106B/206B Guest Lecture Feb 2021

	<i>“Simultaneous Localization and Mapping: A Unifying Optimization-Based Framework.” (with Amay Saxena)</i>	
	UC Berkeley Semiautonomous Seminar <i>“Adversarial-to-Cooperative Games.”</i>	Jun 2020
INVITED POSTER PRESENTATIONS	2025 <i>“Robustly Constrained Dynamic Games with Uncertain Dynamics.”</i>	Oct 2025
TEACHING EXPERIENCE (GEORGIA TECH)	6-Lecture Short Course (Game Theory in Controls) <i>Instructor for a 6-lecture Short Course</i>	Jul 2025 - Aug 2025
TEACHING EXPERIENCE (UC BERKELEY)	EE 127 (Optimization Models in Engineering) <i>20-hour Graduate Student Instructor for a 16-week semester-long course</i>	Aug 2023 - Dec 2023
	EE 221A (Linear Systems Theory) <i>20-hour Graduate Student Instructor for a 16-week semester-long course</i>	Aug 2021 - Dec 2021
	EE 16B (Designing Information Devices and Systems II) <i>25-hour Graduate Student Instructor for an 8-week summer course</i>	Jul 2020 - Aug 2020
JOURNAL AND CONFERENCE REVIEWING	Journals • IEEE Transactions on Networked Control Systems, 2025. • IEEE Transactions on Automatic Control, 2024. • IEEE Control Systems Letters, 2024. • IEEE Robotics and Automation Letters (RA-L), 2024. • Journal of Machine Learning Research (JMLR), 2023. Conferences • IEEE International Conference on Intelligent Transportation Systems, 2024. • IEEE Conference on Decision and Control (CDC), 2024. • IFAC American Control Conference (ACC), 2024. • IEEE International Conference on Robotics and Automation (ICRA), 2022. • IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS), 2021. • IEEE International Conference on Robotics and Automation (ICRA), 2021. • IEEE Conference on Decision and Control (CDC), 2020.	
SOFTWARE SKILLS	Python, Matlab, C++, L ^A T _E X	
SERVICE AND OUTREACH	Graduate Student Recruitment Committee (GSRC) <i>Committee for evaluating Ph.D. applications to Georgia Tech ECE and recommending applicants for first-year Graduate Teaching Assistantships (GTA)</i>	December 2025 - now
	ACC 2025: Workshop on Mixed-Autonomy Traffic, Co-Organizers <i>Workshop on Emerging Challenges and Opportunities in Mixed Autonomy Transportation Systems</i> (https://sites.google.com/view/acc2025workshoponmixedautonomy/home?authuser=0)	July 2024

Decision and Control Laboratory Seminar Co-Organizer

Sep 2024 - now

Regular seminar series focused on emerging challenges in the control of robotic and cyber-physical systems, at Georgia Tech ECE

DREAM Seminar Co-Organizer

Sep 2022 - Dec 2023

Weekly seminar on control theory, robotics, optimization, computer vision, and machine learning, at UC Berkeley EECS.

Semiautonomous Seminar Co-Organizer

Jan 2020 - Dec 2021

Weekly seminar on control theory, robotics, optimization, computer vision, and machine learning, at UC Berkeley EECS.